Solving the Rubik's Cube with A*

Artificial Intelligence and Reasoning, Lab 1

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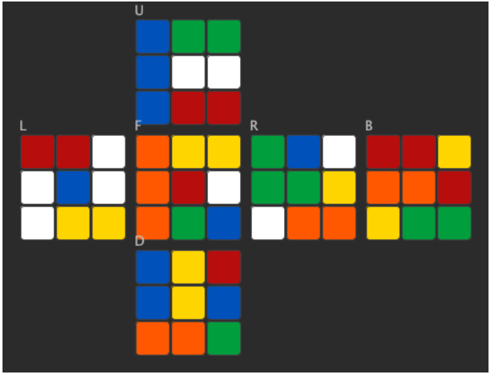
The problem

- A Rubik's Cube has six faces of nine stickers.
- Each face can turn, which mixes the colors.
- The cube is solved when each face is one color.
- It has about 4.3×10^{19} configurations.

Our goal: write the solve button.
Find a list of moves back to the solved cube.
We use search, as seen in Lecture 2.



A scrambled cube, drawn as a net.

The cube as a search problem

We use the five components of Lecture 2.

- **State:** a cube configuration.
- **Actions:** the twelve quarter turns.
- **Transition:** a turn permutes the stickers.
- **Goal test:** every face is uniform.
- **Cost:** each move costs one.

The shortest move list is the optimal solution.
The branching factor is twelve, so the tree is $O(12^d)$.
A blind search cannot go deep. We need a heuristic.

The classes

We followed the statement closely, in Java.

- **Cube:** faces, the twelve turns, isSolved.
- **State:** a node, with g, h and expand.
- **Astar:** the solve method.
- **StateSet:** the explored set.
- **PriorityQueueState:** the frontier.
- **Heuristic:** the interface and its two versions.

The turns come from a real 3D model of the cube.
We built each class test first, with twenty four tests.

The A* algorithm

A* expands the node with the smallest value

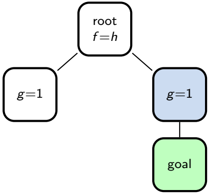
$$f(n) = g(n) + h(n).$$

- $g(n)$ is the cost from the start.
- $h(n)$ estimates the cost to the goal.
- An admissible h never overestimates the cost.
- With an admissible h, A* is optimal.
- With $h = 0$, A* is a uniform cost search.

We rebuild the plan by walking the parent chain.

Uniform cost search and A*

- UCS orders the frontier by g only.
- It is optimal but blind.
- A* adds the heuristic h.
- It looks toward the goal.
- So A* expands far fewer states.
- Both find the same optimal plan.



Smallest f is expanded first.

Heuristic 1: the sticker relaxation

A relaxation drops some rules of the problem.
Here each sticker may move on its own.

- Each sticker must reach its correct face.
- Distance is 0, 1 or 2 quarter turns.
- We sum over all 54 stickers.
- We divide by 12 stickers moved per turn.

$$h_{\text{label}} = \frac{1}{12} \sum_{s=1}^{54} d(s).$$

After one turn, the sum is 12, so $h = 1$.
This is admissible, so A* stays optimal.

Heuristic 2: the piece relaxation

Here each piece may move on its own.

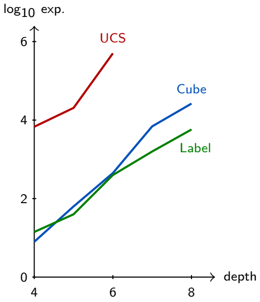
- A piece is a corner or an edge.
- One turn moves four corners and four edges.
- We count the misplaced pieces.
- The centers are fixed, so we ignore them.

$$h_{\text{cube}} = \max\left(\left\lceil \frac{\text{misplaced corners}}{4} \right\rceil, \left\lceil \frac{\text{misplaced edges}}{4} \right\rceil\right).$$

This is also admissible. The sticker one is stronger here.

Results

- UCS blows up around depth seven.
- At depth six it expands about 500k states.
- Both heuristics scale much further.
- Plans stay optimal, same length.
- A* solves depth ten in about twelve seconds.
- Memory is the first limit, not time.



Expanded states by depth, log scale.

Conclusion

- We solved the cube with A*, as in the course.
- A* returns an optimal solution.
- A good heuristic gives huge savings.
- The heuristic changes the effort, not the answer.
- UCS fails from about seven moves.
- The main limit of A* is memory.

Thank you for your attention.